

ORLANDO INTL, FL, USA

WMO: 722050

Lat: 28.434N

Lon: 81.325W

Elev: 90

StdP: 14.65

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12815

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.4	42.4	20.7	15.6	47.8	26.3	20.3	50.4	23.7	61.1	20.8	61.3	7.7	350	0.379

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.9	93.7	76.6	92.4	76.3	91.1	76.0	79.7	87.1	78.9	86.4	78.3	85.6	7.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	145.7	81.6	77.2	142.1	81.0	76.5	138.8	80.7	43.0	87.2	42.3	86.5	41.7	85.9	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	18.2	16.5	DB	31.9	96.5	4.0	1.6	29.1	97.7	26.7	98.6	24.5	99.5	21.6	100.7	(4)
(5)			WB	28.3	81.3	3.7	1.2	25.6	82.1	23.5	82.8	21.4	83.5	18.7	84.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.1	60.5	63.9	67.5	72.4	77.6	81.2	82.5	82.6	81.0	75.6	68.3	63.9
	DBStd	9.44	8.16	7.69	6.63	4.94	3.52	2.58	2.04	1.99	2.21	5.15	6.17	7.96
	HDD50	27	15	5	1	0	0	0	0	0	0	0	0	6
	HDD65	512	184	103	54	8	0	0	0	0	0	5	39	119
	CDD50	8468	340	393	543	673	855	935	1008	1011	929	795	549	437
	CDD65	3480	44	72	132	231	390	485	543	546	479	335	138	85
	CDH74	29387	253	471	946	1855	3440	4323	5182	5193	4140	2417	751	416
CDH80	10814	17	77	200	573	1397	1842	2197	2100	1536	709	121	45	
Wind	WSAvg	7.5	8.0	8.2	8.8	8.5	7.8	6.6	5.9	6.1	6.9	7.8	7.9	7.6
Precipitation	PrecAvg	48.9	2.4	2.5	3.3	2.0	3.7	7.4	7.4	6.9	6.1	2.8	1.9	2.2
	PrecMax	67.9	7.2	8.7	12.7	9.1	14.6	17.4	18.3	15.9	14.1	10.2	10.3	12.6
	PrecMin	30.4	0.1	0.2	0.0	0.0	0.4	1.6	2.6	2.8	1.7	0.0	0.0	0.2
	PrecStd	8.4	1.7	1.9	2.8	1.9	2.7	3.2	3.1	2.8	2.6	2.2	2.1	2.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.1	85.2	87.1	90.9	93.6	96.2	95.1	94.8	92.6	90.3	86.1	83.7
		MCWB	69.2	69.9	70.4	71.8	72.9	76.1	77.5	77.7	76.2	75.9	73.2	71.5
	2%	DB	79.6	82.4	84.6	88.0	91.4	93.5	93.3	93.0	91.0	88.2	83.4	81.2
		MCWB	67.8	68.5	68.9	70.9	72.5	76.3	77.1	77.4	76.0	74.9	71.6	70.2
	5%	DB	77.1	80.0	82.3	85.9	89.5	91.4	91.8	91.5	89.6	86.3	80.9	78.9
		MCWB	66.5	67.3	67.4	69.7	72.1	75.7	77.0	77.3	75.8	73.9	69.8	69.0
	10%	DB	73.8	76.7	79.6	83.5	87.5	89.3	90.1	89.8	88.0	84.2	78.5	76.0
		MCWB	65.2	65.9	66.3	68.4	71.7	75.5	76.9	77.2	75.8	72.9	68.2	67.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	72.4	74.0	75.5	77.3	80.0	80.5	80.5	79.8	78.8	75.5	73.6
		MCDB	77.7	79.4	81.2	83.9	86.3	88.4	87.7	88.0	86.1	84.2	81.1	79.8
	2%	WB	69.9	70.8	72.0	73.7	75.9	78.4	79.5	79.6	78.5	77.6	74.1	72.2
		MCDB	75.7	77.7	79.2	82.0	84.5	86.8	87.1	87.3	85.0	83.2	79.3	77.6
	5%	WB	68.5	69.5	70.5	72.4	75.0	77.8	78.7	78.9	78.0	76.7	72.6	70.8
		MCDB	74.2	76.3	77.7	80.6	83.4	86.1	86.3	86.4	84.5	82.2	77.6	75.7
	10%	WB	66.8	68.1	68.9	71.1	74.0	77.1	78.1	78.3	77.3	75.6	71.0	69.3
		MCDB	72.1	74.6	76.1	79.3	82.2	85.1	85.7	85.5	83.8	80.9	75.9	74.1
Mean Daily Temperature Range	5% DB	MDBR	20.4	20.5	20.9	20.8	19.8	17.1	16.6	15.9	15.0	16.4	18.4	18.7
		MCDBR	21.4	21.9	22.3	22.3	22.0	19.8	18.2	17.6	16.6	16.7	18.6	18.9
		MCWBR	10.8	9.4	8.7	7.7	6.5	6.0	5.4	5.1	5.1	6.0	8.2	9.1
	5% WB	MCDBR	19.1	19.2	19.1	19.7	18.7	16.9	16.6	15.8	14.7	14.1	15.9	16.6
		MCWBR	11.1	9.6	9.2	8.2	6.4	6.1	5.7	5.4	5.3	5.7	8.3	9.6
Clear-Sky Solar Irradiance	taub	0.339	0.353	0.373	0.391	0.425	0.454	0.475	0.463	0.422	0.397	0.358	0.346	
	taud	2.524	2.496	2.429	2.357	2.309	2.300	2.239	2.289	2.398	2.431	2.495	2.526	
	Ebn at Noon	287	292	291	287	276	266	260	263	272	272	277	279	
	Edh at Noon	29	32	36	40	42	42	44	42	36	33	29	27	
All-Sky Solar Radiation	RadAvg	1088	1289	1630	1929	1999	1795	1796	1709	1537	1401	1153	992	
	RadStd	80	82	121	97	135	166	106	113	86	97	61	77	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-81	+243	+180
Regional (41 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-9	-85	+284	+203

Nomenclature: See separate page